

第一题: (20 分)

量子力学 2021
725 1~6

- 1) 不考虑相对论, 电子经过高电压 V 加速, 求德布罗意波长;
- 2) 写出坐标表象下 x 和 p 的表达式, 求其对易关系 $[x, p]$;
- 3) 一维体系, 当势函数为偶函数即 $V(x) = V(-x)$ 时, 求证波

函数总有确定的宇称;

解: (1) 德布罗意波长: $\lambda = \frac{h}{p}$

平均动能: $\frac{p^2}{2m} = qU$ $\therefore \lambda = \frac{h}{\sqrt{2mqU}}$

(2) $\hat{p} = -i\hbar \frac{\partial}{\partial x}$

$$[x, \hat{p}] \psi(x) = (x\hat{p} - \hat{p}x) \psi(x) = x(-i\hbar \frac{\partial}{\partial x} \psi) + i\hbar \frac{\partial}{\partial x} (x\psi)$$

$$= -i\hbar (x \frac{\partial \psi}{\partial x} - \psi - x \frac{\partial \psi}{\partial x}) = i\hbar \psi(x)$$

$\therefore [x, \hat{p}] = i\hbar$

Δ (3) 当 $V(x) = V(-x)$ 时, 令 $x' = -x$.

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(-x) + V(x) \psi(-x) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x'^2} \psi(x') + \underbrace{V(-x')} \psi(x')$$

为什么选 $\psi(-x)$?

$$= -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x'^2} \psi(x') + \underbrace{V(x')} \psi(x')$$

$$= E \psi(x') \Rightarrow \text{统一变量才可以相等}$$

$$= E \psi(-x)$$

$\therefore \psi(x)$ 和 $\psi(-x)$ 是同一方程的解.

$\psi(x) \pm \psi(-x)$ 也是同一薛定谔方程的解.

$\therefore$ 波函数有确定宇称.

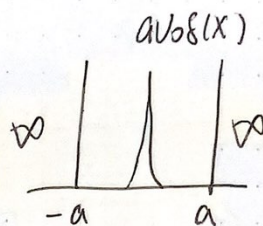
$\hat{p} \psi(x) = \psi(-x)$, 波函数为偶宇称.

$\hat{p} \psi(x) = -\psi(-x)$, 波函数为奇宇称.

$\hat{p}$ 为宇称算符.

第二题: (30 分)

一无限深方势阱两边坐标为 $-a$ 和 a , 存在一 δ 势垒, $V(x) = aV_0\delta(x)$ 试根据以下步骤求解:



- 1) 写出薛定谔方程, 及波函数的表达式
- 2) 写出波函数在 $\pm a$ 处的边界条件以及在 $x=0$ 处的跳跃条件;
- 3) 势能具有对称性, 故波函数有确定的宇称。试求奇宇称的波函数及其能级, 与无限深方势阱的能级对比;
- 4) 求偶宇称解的能级满足的公式, 通过画图法求解本征能量, 并讨论当 a 很大或者很小的情况, 对比奇宇称能级, 说明哪一个

→ 套公式

解: (1) $-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + V(x)\psi(x) = E\psi(x)$

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + aV_0\delta(x)\psi(x) = E\psi(x)$$

无限深方势阱: $\psi(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$ 宽度 $0 \sim a$.

宽度 $0 \sim 2a$. 波函数: $\psi(x) = \sqrt{\frac{1}{a}} \sin \frac{n\pi(x+a)}{2a}$

(2) 边界条件 $\psi(\pm a) = 0$.

跳跃条件: $\psi'(0^+) - \psi'(0^-) = \frac{2maV_0}{\hbar^2} \psi(0)$

(3) 奇宇称: $\psi(0) = 0$.

当 $x \neq 0$ 时, Sch. equ: $\delta(x) = 0$.

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) = E\psi(x) \quad \text{令 } k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\therefore \frac{d^2}{dx^2} \psi(x) + k^2 \psi(x) = 0$$

一般解: $\psi(x) = \begin{cases} A \sin kx + B \cos kx & 0 < x < a \\ C \sin kx + D \cos kx & -a < x < 0 \end{cases}$

边界条件: $\psi(a) = \psi(-a) = 0$. $\psi(0^+) = \psi(0^-) = 0$.

$B = D = 0$. $A = C$.

$$\therefore \psi(0^+) - \psi(0^-) = \frac{2ma}{\hbar^2} \psi(0)$$

$$\psi(x) = A \sin kx$$

$$\sin ka \neq 0 \quad ka = n\pi \quad k = \frac{n\pi}{a} \quad n=1, 2, 3, \dots$$

$$\sqrt{\frac{2mE}{\hbar^2}} = \frac{n\pi}{a}$$

$$E = \frac{n^2 \pi^2 \hbar^2}{2m a^2} \quad n=1, 2, 3.$$

$$A^2 \int_{-a}^a \sin^2 kx \, dx = 1$$

$$A^2 \int_{-a}^a \frac{1 - \cos 2kx}{2} \, dx = 1$$

$$A^2 \left(\frac{1}{2}x - \frac{\sin 2kx}{4k} \right) \Big|_{-a}^a = 1$$

$$A = \sqrt{\frac{1}{a}}$$

$$\psi = \sqrt{\frac{1}{a}} \sin \frac{n\pi x}{a} \quad -a \leq x \leq a.$$

无限深方势阱:

$$\psi_k = \sqrt{\frac{1}{a}} \sin \frac{k\pi x}{a}$$

$$E_k = \frac{k^2 \pi^2 \hbar^2}{8ma^2}$$

$$\therefore k^2 = 4n^2. \therefore k = 2n.$$

$\therefore$ 取无限深方势阱的奇项

(4) 偶宇称 $x \neq 0$ 时, $V=0$

$$\frac{d^2}{dx^2} \psi(x) + k^2 \psi(x) = 0.$$

$$\psi(x) = \begin{cases} A \cos kx + B \sin kx & 0 < x < a \\ A \cos kx - B \sin kx & -a < x < 0 \end{cases}$$

$$\text{阶跃条件: } \psi'(0^+) - \psi'(0^-) = \frac{2m\alpha V_0}{\hbar^2} \psi(0)$$

$$Bk + Bk = \frac{2m\alpha V_0}{\hbar^2} A.$$

$$B = \frac{m\alpha V_0}{\hbar^2 k} A = \lambda A. \quad \text{令 } \frac{m\alpha V_0}{\hbar^2 k} = \lambda$$

$$\therefore \psi(a) = \psi(-a) = 0. \quad A \cos ka + \lambda A \sin ka = 0$$

$$\tan ka = -\frac{1}{\lambda} = -\frac{\hbar^2 k}{m\alpha V_0}.$$

$$-ka \cot ka = \frac{m\alpha V_0 a}{\hbar^2}$$

$$-\xi \cot \xi = \frac{m\alpha V_0 a}{\hbar^2}$$

第三题: (25 分)

一维谐振子, 势能函数为 $V(x) = \frac{1}{2}m\omega^2 x^2$, 升降算符为

$$a_{\pm} = \frac{1}{\sqrt{2\hbar m\omega}} (\mp i p + m\omega x), \text{ 且已知 } a_+ \psi_n = \sqrt{n+1} \psi_{n+1}, a_- \psi_n =$$

$\sqrt{n} \psi_{n-1}$, 充分利用升降算符的性质, 试求

- 1) 若粒子处于量子态 ψ_n , 求位置和动量的平均值 $\langle x \rangle, \langle p \rangle$;
- 2) 若粒子处于量子态 $\psi_n, \langle x^2 \rangle, \langle p^2 \rangle$;
- 3) 求不确定度 $\sigma_x = \langle (x - \bar{x})^2 \rangle, \sigma_p = \langle (p - \bar{p})^2 \rangle$, 并验证不确定度关系, 说明等号什么情况下成立;
- 4) 若粒子处于 $\psi = a\psi_1 + b\psi_2$ 量子态, 求位置的平均值;
- 5) 上问中波函数经过 t 时间的演化后, 求位置的期望。

解: 1) $x = \frac{\sqrt{\hbar}}{\sqrt{2m\omega}} (a_+ + a_-)$ $p = i \sqrt{\frac{\hbar m\omega}{2}} (a_+ - a_-)$

$$\begin{aligned} \langle x \rangle &= \langle \psi_n | x | \psi_n \rangle = \frac{\sqrt{\hbar}}{\sqrt{2m\omega}} \langle \psi_n | a_+ + a_- | \psi_n \rangle \\ &= \frac{\sqrt{\hbar}}{\sqrt{2m\omega}} (\langle \psi_n | \sqrt{n+1} | \psi_{n+1} \rangle + \langle \psi_n | \sqrt{n} | \psi_{n-1} \rangle) = 0 \end{aligned}$$

$$\begin{aligned} \langle p \rangle &= \langle \psi_n | p | \psi_n \rangle = i \sqrt{\frac{\hbar m\omega}{2}} \langle \psi_n | a_+ - a_- | \psi_n \rangle \\ &= i \sqrt{\frac{\hbar m\omega}{2}} (\langle \psi_n | \sqrt{n+1} | \psi_{n+1} \rangle - \langle \psi_n | \sqrt{n} | \psi_{n-1} \rangle) = 0 \end{aligned}$$

$$\begin{aligned} (2) \quad \langle x^2 \rangle &= \langle \psi_n | \frac{\hbar}{2m\omega} (a_+ + a_-)^2 | \psi_n \rangle = \frac{\hbar}{2m\omega} \langle \psi_n | a_+^2 + a_-^2 + a_+ a_- + a_- a_+ | \psi_n \rangle \\ &= \frac{\hbar}{2m\omega} \langle \psi_n | a_+ a_- + a_- a_+ | \psi_n \rangle = \frac{\hbar}{2m\omega} \langle \psi_n | a_+ a_- + a_+ a_- + 1 | \psi_n \rangle \\ &= \frac{\hbar}{2m\omega} \langle \psi_n | 2n+1 | \psi_n \rangle = \frac{\hbar}{2m\omega} (2n+1) \end{aligned}$$

$$\begin{aligned} \langle p^2 \rangle &= \langle \psi_n | \frac{\hbar m\omega}{2} (a_+^2 + a_-^2 - a_+ a_- - a_- a_+) | \psi_n \rangle \\ &= \langle \psi_n | \frac{\hbar m\omega}{2} (a_+ a_- + a_- a_+) | \psi_n \rangle = \frac{\hbar m\omega}{2} (2n+1) \end{aligned}$$

$$(3) \quad \sigma_x = \langle (x - \langle x \rangle)^2 \rangle = \langle x^2 - 2x\langle x \rangle + \langle x \rangle^2 \rangle = \frac{\hbar}{2m\omega} (2n+1)$$

$$\sigma_p = \langle (p - \langle p \rangle)^2 \rangle = \langle p^2 - 2p\langle p \rangle + \langle p \rangle^2 \rangle = \frac{\hbar m\omega}{2} (2n+1)$$

$$\sigma_x \sigma_p = \frac{\hbar^2}{4} (2n+1)^2 \geq \frac{\hbar^2}{4} \text{ 当且仅当 } n=0 \text{ 粒子处于基态 } \psi_0 \text{ 时成立}$$

$$\begin{aligned}
 (4) \quad \langle X \rangle &= \langle \psi | X | \psi \rangle \\
 &= \langle a\psi_1 + b\psi_2 | \frac{\hbar}{2m\omega} (a + a_-) | a\psi_1 + b\psi_2 \rangle \\
 &= \frac{\hbar}{2m\omega} [\langle a\psi_1 + b\psi_2 | a | a\psi_1 + b\psi_2 \rangle + \langle a\psi_1 + b\psi_2 | a_- | a\psi_1 + b\psi_2 \rangle] \\
 &= \frac{\hbar}{2m\omega} [\langle a\psi_1 + b\psi_2 | (\sqrt{2}a|\psi_2\rangle + \sqrt{2}b|\psi_3\rangle) + \langle a\psi_1 + b\psi_2 | (a|\psi_0\rangle + \sqrt{2}b|\psi_1\rangle)] \\
 &= \frac{\hbar}{2m\omega} (\sqrt{2}b^*a + \sqrt{2}a^*b)
 \end{aligned}$$

$$(5) \quad E_n = (n + \frac{1}{2})\hbar\omega \quad E_1 = \frac{3}{2}\hbar\omega \quad E_2 = \frac{5}{2}\hbar\omega.$$

$$\begin{aligned}
 \psi(t) &= a e^{-iE_1 t/\hbar} \psi_1 + b e^{-iE_2 t/\hbar} \psi_2 \\
 &= a e^{-i3\omega t/2} \psi_1 + b e^{-i5\omega t/2} \psi_2
 \end{aligned}$$

$$\langle X(t) \rangle = \langle \psi(t) | X | \psi(t) \rangle$$

$$\frac{\hbar}{2m\omega} \langle a e^{+i3\omega t/2} \psi_1 + b e^{+i5\omega t/2} \psi_2 | (\sqrt{2} a e^{-i3\omega t/2} \psi_2 + \sqrt{2} b e^{-i5\omega t/2} \psi_3) \rangle$$

$$\frac{\hbar}{2m\omega} \langle a e^{+i3\omega t/2} \psi_1 + b e^{+i5\omega t/2} \psi_2 | (a e^{-i3\omega t/2} \psi_0 + \sqrt{2} b e^{-i5\omega t/2} \psi_1) \rangle$$

$$\frac{\hbar}{2m\omega} (\sqrt{2}b^*a e^{+i\omega t} + \sqrt{2}a^*b e^{-i\omega t}).$$

第四题: (30 分)

$$\psi_{nlm} \quad n=1 \quad n=2$$

已知一群氢原子处于波函数 $\Psi(r, t) = A\psi_{100}e^{-\frac{r}{a}} + B\psi_{210}e^{-\frac{r}{2a}} +$

$C\psi_{211}e^{-\frac{r}{2a}}$ 所表示的量子态中, 其中 A、B、C 为复数, 且有

$$\psi_{100} = \frac{1}{\sqrt{\pi a^3}} e^{-r/2a}$$

$$\psi_{210} = \dots e^{-r/2a} r \cos \theta \quad (\text{前面的系数忘了, 但是系数不重要})$$

$$\psi_{211} = \dots e^{-r/2a} r \sin \theta e^{i\phi}$$

现在想要通过一些列测量来确定氢原子的量子态, 试求:

1) 能量的可能测值及相应概率;

2) $t=0$ 时刻, L^2 和 L_z 的可能测值及其概率;

3) 在 t 时刻, 求 L_x 的可能测值及其概率;

4) 位置分量 x, y, z 的平均值;

5) 测定了上述力学量后, 可以完全确定氢原子的量子态了吗? 说明原因 $\rightarrow$ 不行

1) 氢原子: $E_n = -\frac{e^2}{2an^2}$

可能值: $E_1 = -\frac{e^2}{2a} \quad E_2 = -\frac{e^2}{8a}$

概率: $P_1 = |A|^2 \quad P_2 = |B|^2 + |C|^2$

(2) $t=0$ 时: $\psi(0) = A\psi_{100} + B\psi_{210} + C\psi_{211}$

$\hat{L}^2$ 可能值: $\hat{L}_1^2 = l(l+1)\hbar^2 = 0 \quad \hat{L}_2^2 = l(l+1)\hbar^2 = 2\hbar^2$

概率 $P_1 = |A|^2 \quad P_2 = |B|^2 + |C|^2$

L_z 可能值: $\hat{L}_{z1} = m\hbar = 0 \quad \hat{L}_{z2} = m\hbar = 1$

概率 $P_1 = |A|^2 + |B|^2 \quad P_2 = |C|^2$

(3) $\hat{L}_x = \frac{1}{2}(\hat{L}_+ + \hat{L}_-)$ $\hat{L}_{\pm}|\psi_{nlm}\rangle = \sqrt{l(l+1) - m(m\pm 1)}\hbar|\psi_{nlm\pm 1}\rangle$

$\hat{L}_x|\psi(r, t)\rangle = \frac{1}{2}(\hat{L}_+ + \hat{L}_-)|\psi(r, t)\rangle = \frac{1}{2}(\sqrt{2}\hbar \cdot B\psi_{211}e^{iEt/\hbar} + \sqrt{2}\hbar \cdot C\psi_{210}e^{iEt/\hbar})$

$\therefore L_x$ 可能测量值为 0. 概率 1

(4) $x = r \sin \theta \cos \phi$

$y = r \sin \theta \sin \phi$

$z = r \cos \theta$

$\langle x \rangle = \langle \psi(r, t) | x | \psi(r, t) \rangle$

$= |A|^2 \langle \psi_{100} | x | \psi_{100} \rangle + |B|^2 \langle \psi_{210} | x | \psi_{210} \rangle$

$+ |C|^2 \langle \psi_{211} | x | \psi_{211} \rangle$

$= 0$

同理 $\langle y \rangle = 0 \quad \langle z \rangle = 0$ $\rightarrow$ 草稿纸上算

第五题: (25 分)

空间中有 x 方向均匀磁场, 大小为 B , 质量 m 电量 q 自旋为 $1/2$ 的电子在其中运动, 开始时电子处于自旋向上状态 (S_z 本征值为 $\hbar/2$),

试求:

$$|\uparrow\rangle$$

- 6.8 {
- 1) t 时刻电子的波函数;
 - 2) t 时刻电子自旋 y 分量的可能测值及概率;
 - 3) 电子自旋第一次反向的时间 T ;
 - 4) 在 $t = 2T$ 时刻, 对电子进行测量, 发现其处于自旋 x 分量本征值为 $\hbar/2$ 的本征态, 求 $t = 3T$ 时刻电子自旋 x 分量的可能测值及其概率

解: (1) $\vec{H} = -\vec{m}_s \cdot \vec{B} = -\frac{q}{mc} \vec{S} \cdot \vec{B} = -\frac{qB}{mc} \hat{S}_x = -\frac{qB\hbar}{2mc} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$|\psi(0)\rangle = |\uparrow\rangle$$

$$|\psi(t)\rangle = e^{-i\vec{H}t/\hbar} |\psi(0)\rangle = e^{i\frac{qB}{mc} \hat{S}_x t/\hbar} |\uparrow\rangle$$

用 $|\uparrow\rangle$ 将 S_x 的本征函数展开.

$$|\uparrow\rangle = (|\psi_+\rangle\langle\psi_+| + |\psi_-\rangle\langle\psi_-|) |\uparrow\rangle$$

$$= |\psi_+\rangle\langle\psi_+|\uparrow\rangle + |\psi_-\rangle\langle\psi_-|\uparrow\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} |\psi_+\rangle + \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \end{bmatrix} |\psi_-\rangle = \frac{1}{\sqrt{2}} (|\psi_+\rangle + |\psi_-\rangle)$$

$$|\psi(t)\rangle = e^{i\frac{qB}{mc} \hat{S}_x t/\hbar} \frac{1}{\sqrt{2}} (|\psi_+\rangle + |\psi_-\rangle)$$

$$= \frac{1}{\sqrt{2}} \left(e^{i\frac{qB}{mc} \frac{\hbar}{2} t/\hbar} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + e^{-i\frac{qB}{mc} \frac{\hbar}{2} t/\hbar} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right)$$

$$= \frac{1}{2} \begin{bmatrix} e^{iqBt/2mc} + e^{-iqBt/2mc} \\ e^{iqBt/2mc} - e^{-iqBt/2mc} \end{bmatrix} = \begin{bmatrix} \cos \frac{qBt}{2mc} \\ i \sin \frac{qBt}{2mc} \end{bmatrix}$$

(2) t 时刻, 粒子 $S_y = \frac{\hbar}{2}$ 概率: $\omega = \frac{qB}{mc}$

$$p = |\langle\psi_+|\psi(t)\rangle|^2 = \left| \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i \end{bmatrix} \begin{bmatrix} \cos \omega t \\ i \sin \omega t \end{bmatrix} \right|^2 = \frac{1}{2} (1 + 2 \sin \omega t \cos \omega t) = \frac{1}{2} (1 + \sin 2\omega t)$$

$$\text{粒子 } S_y = -\frac{\hbar}{2} \text{ 概率: } p = |\langle\psi_-|\psi(t)\rangle|^2 = \left| \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \end{bmatrix} \begin{bmatrix} \cos \omega t \\ i \sin \omega t \end{bmatrix} \right|^2 = \frac{1}{2} (1 - \sin 2\omega t)$$

$$(3) \therefore |\psi(t)\rangle = \begin{pmatrix} \cos \omega t \\ i \sin \omega t \end{pmatrix} \quad \omega = \frac{qB}{2mc}$$

$$\psi(0) = |\uparrow\rangle \text{ 则自旋反向 } |\psi(T)\rangle = |\downarrow\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{即 } |\cos \omega t|^2 = 0, \quad |\sin \omega t|^2 = 1$$

$$\therefore \omega t = (n + \frac{1}{2})\pi \quad n = 0, 1, 2, \dots$$

$$\text{第1次反向即 } \omega T = \frac{\pi}{2} \quad \therefore T = \frac{\pi}{\omega} = \frac{\pi mc}{qB}$$

$$(4) \quad |\psi(t=2T)\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \cos(\omega 2T) \\ i \sin(\omega 2T) \end{bmatrix}$$

$$|\cos 2\omega T|^2 = \frac{1}{2} \quad |\sin 2\omega T|^2 = \frac{1}{2} \quad \therefore 2\omega T = (n + \frac{1}{4})\pi$$

$$|\psi(t=3T)\rangle = \begin{bmatrix} \cos(\omega \cdot 3 \cdot \frac{\pi}{8\omega}) \\ i \sin(\omega \cdot 3 \cdot \frac{\pi}{8\omega}) \end{bmatrix}$$

$$T = \frac{\pi}{8\omega} = \frac{\pi mc}{4qB}$$

$$= \begin{bmatrix} \cos \frac{3\pi}{8} \\ i \sin \frac{3\pi}{8} \end{bmatrix}$$

$$S_x = \frac{\hbar}{2} \quad |\psi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad P = |\langle \psi_1 | \psi(t=3T) \rangle|^2 = \left| \frac{1}{\sqrt{2}} (1 \ 1) \begin{bmatrix} \cos \frac{3\pi}{8} \\ i \sin \frac{3\pi}{8} \end{bmatrix} \right|^2$$

$$= \frac{1}{2} (\cos \frac{3\pi}{8} - i \sin \frac{3\pi}{8}) (\cos \frac{3\pi}{8} + i \sin \frac{3\pi}{8})$$

$$= \frac{1}{2}$$

$$S = -\frac{\hbar}{2} \quad P = |\langle \psi_2 | \psi(t=3T) \rangle|^2 = \left| \frac{1}{\sqrt{2}} (1 \ -1) \begin{bmatrix} \cos \frac{3\pi}{8} \\ i \sin \frac{3\pi}{8} \end{bmatrix} \right|^2 = \frac{1}{2}$$

第六题: (20 分)

一维无限深势阱两端坐标为 0 和 a , 有两个全同粒子在其中运动, 试求:

- 1) 若粒子自旋为 0, 求体系基态和第一激发态的能量和波函数;
- 2) 若粒子自旋为 $1/2$, 求体系基态和第一激发态的能量和波函数;
- 3) 粒子自旋为 0, 求在微扰势 $V(x) = -aV_0\delta(x_2 - x_1)$ 作用下, 体系基态和第一激发态能量一级修正

解: (1) $S=0$ 为玻色子 满足交换对称

一维无限深势阱: $\psi_n = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$ $E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$ $n=1, 2, 3, \dots$

基态: $\psi_1 = \psi_1(x_1) \psi_1(x_2)$ $E_1 = \frac{\pi^2 \hbar^2}{2ma^2} \times 2 = \frac{\pi^2 \hbar^2}{ma^2}$

第一激发态: $\psi_2 = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) + \psi_2(x_1) \psi_1(x_2)]$

$$E_2 = E_1 + E_2 = \frac{\pi^2 \hbar^2}{2ma^2} + \frac{4\pi^2 \hbar^2}{2ma^2} = \frac{5\pi^2 \hbar^2}{2ma^2}$$

(2) $S=\frac{1}{2}$ 费米子 整体波函数满足交换反对称性.

基态: $\psi_1 = \psi_1(x_1) \psi_1(x_2) |00\rangle$ $E_1 = E_1 + E_1 = \frac{\pi^2 \hbar^2}{ma^2}$

第一激发态: $\psi_2 = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) + \psi_2(x_1) \psi_1(x_2)] |00\rangle$

$$\psi_3 = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) - \psi_2(x_1) \psi_1(x_2)] \begin{cases} |11\rangle \\ |10\rangle \\ |1-1\rangle \end{cases}$$

三重简并.

$$E_3 = E_1 + E_2 = \frac{5\pi^2 \hbar^2}{2ma^2}$$

(3) $H' = -aV_0\delta(x_2 - x_1)$ 由于 $S=0$, 能量非简并.

$$E_1^{(1)} = \hat{H}'_{11} = \langle \psi_1 | \hat{H}' | \psi_1 \rangle = \langle \psi_1(x_1) \psi_1(x_2) | -aV_0\delta(x_2 - x_1) | \psi_1(x_1) \psi_1(x_2) \rangle$$

$$= -aV_0 \int_0^a \psi_1^2(x) \psi_1^2(x) dx = -aV_0 \int_0^a |\psi_1^2(x)|^2 dx$$

$$= -aV_0 \int_0^a \left(\frac{2}{a}\right)^2 \sin^4 \frac{\pi x}{a} dx = -aV_0 \left(\frac{2}{a}\right)^2 \cdot \frac{3}{8} a = -\frac{3}{2} \frac{V_0 \pi}{a}$$

第一激发态: $\psi_{II} = \frac{1}{\sqrt{2}} [\psi_1(x_1)\psi_2(x_2) + \psi_2(x_1)\psi_1(x_2)]$

$$E_2^{(1)} = H_{22}' = -\frac{aV_0}{2} \int_0^a [\psi_1^*(x_1)\psi_2^*(x_2) + \psi_2^*(x_1)\psi_1^*(x_2)] \delta(x_2 - x_1)$$

$$[\psi_1(x_1)\psi_2(x_2) + \psi_2(x_1)\psi_1(x_2)] dx$$

$$= -\frac{aV_0}{2} \int_0^a [|\psi_1(x_1)\psi_2(x_1)|^2 + |\psi_2(x_1)\psi_1(x_1)|^2 +$$

$$+ \psi_1^*(x_1)\psi_2^*(x_1)\psi_2(x_1)\psi_1(x_1) + \psi_2^*(x_1)\psi_1^*(x_1)\psi_1(x_1)\psi_2(x_1)] dx$$

$$= -\frac{aV_0}{2} \times 4 \int_0^a (\psi_1(x_1)\psi_2(x_1))^2 dx$$

$$= -2aV_0 \int_0^a \frac{4}{a^2} \sin \frac{\pi x}{a} \sin \frac{2\pi x}{a} dx$$

$$= \frac{4V_0}{a} \int_0^a (\cos \frac{3\pi x}{a} - \cos \frac{\pi x}{a}) dx = 0.$$